

Async Research Assistant

Bailar: CLI, deployment and assembly

AI-ENG-110 / AI Academy / Topic 4

Prepared 17 September 2026. Team: Nural, Emil, Mahammadali and Bailar. This report completes Bailar's sections and records integration evidence. The other members' authored report sections and six slides have not been supplied. This is a contribution report, not a declaration that the final team release is ready.

Command-line interface

The entry point is `python -m researcher`. `Argparse` exposes `ask`, `demo` and `bench`. Questions must contain 3 to 2000 characters after trimming. The CLI delegates model validation to `ResearchRequest`. The `wiki` alias becomes `wikipedia`; invalid source names and source limits exit with status 2. The accepted source limit is 1 to 10. The core applies its configured cap.

`ask` renders the question, answer, numbered references with URLs and optional warnings. JSON output uses `AnswerWithCitations.to_dict()` plus `timings_ms`, so citations contain index, title, url and origin. Logs go to `stderr` while JSON stays on `stdout`. Expected operational errors exit 1 with a short message.

`--offline` supplies canned sources and a fake synthesizer, requiring no API keys. Its answer labels the demonstration explicitly. Synthetic arXiv/web references use `example.test` rather than misrepresenting a real publication. `--no-cache` uses a null store without constructing SQLite. The service adapter forwards `use_cache=False` and absorbs writes through the null store.

CLI unit tests replace `Researcher` with a fake. They verify argument failures, alias forwarding, cache bypass, cache closure after success and failure, JSON, citations, warning rendering and `demo/benchmark` dispatch. Separate tests exercise the local `OfflineService` with HTTP calls forbidden.

Docker and continuous integration

The two stages use `python:3.12.6-slim`. The builder installs the existing pinned requirements into `/opt/venv`. The runtime copies that environment and runs as `appuser`. `/app` is owned by `appuser` so SQLite, artefacts and benchmark output can be written. The default command displays `ask help`.

The Docker context excludes environment files, databases, logs, caches and `report/slide` build outputs. Offline runs need neither `.env` nor credentials. For live runs, pass an existing private `.env` file at runtime.

CI runs lint, type checking, `pytest` with a 60% coverage threshold, and a Docker build on pull requests. Tools come from `requirements.txt` without unpinned upgrades. The Docker job checks help, a non-root UID and all three offline commands with `--network none`. Repository branch protection must require these jobs for failures to block merging; editing YAML alone cannot enforce that.

Docker is unavailable on this development machine. Image build/run and the required second-machine validation remain unverified until CI or a teammate with Docker executes them. No passing Docker result is claimed here.

Measured verification

Python 3.12.10 on Windows 11 build 26200, AMD64, 12 logical CPUs. All 86 offline tests passed. researcher coverage: 93.04%. CLI coverage: 94%. The CLI and AI smoke subset passed 43 tests. Scoped Ruff checks passed. Scoped mypy with --follow-imports=silent passed for researcher/cli.py.

The normal suite encountered Windows access errors creating restricted pytest temporary folders. The verification launcher changed only os.mkdir's mode to 0o777 inside the test process and used a workspace temporary directory. Test assertions and repository files were unchanged. This environment workaround must not be mistaken for verification on another operating system.

Mahammadali's benchmark ran unchanged with output and SQLite paths redirected to scratch files. Five questions used three sources and 300 ms simulated I/O per source, with a cleared cache and semaphore 3. Sequential: 4.661 seconds. Concurrent: 3.202 seconds. Speedup: 1.46x. See benchmark.md for captured details. The run measures fetching, not live providers or LLM synthesis. Client setup and scheduling remain in the measured concurrent path. One run is not an SLA. The unchanged demo also completed all five offline questions in scratch output.

Integration handoff

Global mypy reports three pre-existing errors: researcher/config.py:42 (_Env redefinition), researcher/logging_setup.py:16 (optional string), and ai/providers/openai.py:143 (optional dictionary key). Global Ruff reports 16 issues in scripts/bench.py, scripts/demo.py, demo_ai.py and tests/test_ai_smoke.py. These files belong to other owners or are frozen and remain unchanged.

Full requirements installation on this Windows machine stopped while building pyreqwest-impersonate, required by duckduckgo-search==6.1.0, because the build needs Rust. Offline checks used the pinned core/testing packages. No dependency versions changed. Live providers and full installation remain unverified here.

researcher.db is already tracked on main, contrary to the final-release rule. The team must coordinate its removal. Other owners must resolve their lint/type errors and provide their report sections and slides before final assembly. The reviewer assigned to Bailar is Mahammadali. The author merges after approval. Keep each PR below 300 changed lines; submit these commits in separate reviews where needed. Squashing changes the commit-share arithmetic.

AI disclosure and ownership

Codex assisted Bailar with edits to researcher/cli.py, tests/test_cli.py, Dockerfile, .dockerignore, .github/workflows/ci.yml and the Docker/timings sections of README.md, and with these report/slides materials. It ran offline checks and drafted the explanation. Bailar must review and understand the changes before submission. No work is attributed to another member and no other member's implementation files were edited.

Evidence sources

The supplied team ownership rules; docs/architecture.md; researcher/cli.py; tests/test_cli.py; Dockerfile; .github/workflows/ci.yml; scripts/bench.py; scripts/demo.py; and the local test, Ruff and mypy outputs from this session.